

Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS) gene expression signatures for the identification of compounds targeting metabolism

Ankush Dehlia

Deakin University School of Life and Environmental Sciences

Zahra Alizada

Deakin University School of Life and Environmental Sciences

Mark Andrew Guthridge

m.guthridge@deakin.edu.au

Deakin University School of Life and Environmental Sciences <https://orcid.org/0000-0002-0536-3471>

Research Article

Keywords: Myalgic Encephalomyelitis, Chronic Fatigue Syndrome, mitochondria, connectivity mapping, gene signature, PPAR agonist

Posted Date: July 5th, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-10076074/v1>

License:   This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Abstract

Background

Pathological fatigue is a debilitating symptom in a range of chronic illnesses, including myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS). However, drug therapies that are effective at treating pathological fatigue have proved difficult to identify through conventional drug discovery pipelines. We sought to identify an ME/CFS gene expression signature in order to map disease pathways and identify potentially novel therapeutic approaches.

Methods

We analysed ME/CFS patient gene expression profiles obtained from two independent studies (GSE14577 and GSE227375). The identified differentially expressed genes (DEGs) were subjected to CMap L1000 analysis for the identification of functional gene networks as well as compounds with the potential to restore the disrupted gene expression in ME/CFS patients.

Results

We identified 21 up-regulated and 25 down-regulated DEGs common to both gene expression datasets. Subjecting the 46-gene signature to CMap L1000 analysis identified networks involved in cytokine/growth factor signalling, infection and immunity, metabolism and cell growth, lipid metabolism, ubiquitination, nucleotide metabolism, and redox regulation. Four compounds, pioglitazone, rosiglitazone (PPAR agonists), tideglusib (GSK3 inhibitor), and MHY1485 (mTORC1 agonist), were identified and tested for their ability to regulate the metabolism and growth of C2C12 muscle cells. Pioglitazone, tideglusib, and MHY1485, but not rosiglitazone, induced a modest but significant increase in mitochondrial membrane potential in differentiated C2C12 myoblasts over 24–48 hours. These changes occurred in the absence of increased mitochondrial mass and ROS production, suggesting enhanced mitochondrial metabolism rather than mitochondrial biogenesis. While pioglitazone and rosiglitazone increased the growth of C2C12 myoblasts, no significant effects were observed for tideglusib and MHY1485.

Conclusions

Our findings, based on reciprocal disease:drug signature-matching, provide a framework for exploring the therapeutic potential of drugs to target impaired metabolic pathways in conditions involving pathological fatigue, such as ME/CFS.

Background

Approximately 25% of primary care patients report fatigue, indicating that it is an extremely common symptom that is associated with a wide range of pathologies (1, 2). Physiological fatigue after physical exertion is transient and can be relieved by rest and lifestyle management. In contrast, pathological fatigue, which is commonly associated with long-term chronic inflammatory conditions, persists despite adequate rest and lifestyle interventions (3). In prevalent illnesses such as myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), multiple sclerosis (MS), cancer-related fatigue (CRF), post-stroke fatigue (PSF), and more recently long-COVID-19, pathological and persistent fatigue is associated with exercise intolerance, a decline in physical function and poor health-related quality-of-life (4, 5). In part because pathological fatigue associated with illness can be clinically mistaken for physiological tiredness or sleepiness, clinical recognition of its major impacts on daily functioning and longer-term quality of life is often overlooked (6). These limitations may also extend to research where the pathophysiology of fatigue and potential therapeutic targets remain largely unclear for many chronic illnesses (7).

There are a few pharmacological treatments for pathological fatigue, particularly in relation to muscle dysfunction and exercise intolerance. To date, most clinical trials have investigated immunological or psychotropic drugs. For example, in ME/CFS, where fatigue is a primary symptom, two systematic analyses of over 45 trials did not identify any effective drug treatments apart from the possible exceptions of the immunomodulatory drug rintatolimod and the psychostimulant methylphenidate, where modest improvements in a subset of patients were observed (8, 9). In at least some MS patients, the psychostimulant, modafinil, and the dopaminergic agent, amantadine, appear to be the only drugs that provide significant, albeit modest, improvements in fatigue (10, 11). In a systematic analysis of 32 CRF trials, none of the drugs examined, including dexamphetamine, modafinil, methylphenidate or paroxetine, were effective for fatigue (12). Similarly, systematic analysis of 22 PSF trials examining a range of drugs, including modafinil, fluoxetine, and the monoamine stabilizer, OSU6162, reported that none provided significant improvements in fatigue (13). Additionally, a meta-analysis of 45 studies examining palliative care in patients with multiple sclerosis, post-polio syndrome, Parkinson's disease, cancer, HIV/AIDS, end-stage renal disease, multi-type advanced disease in hospice patients, amyotrophic lateral sclerosis, and end-stage chronic lung disease indicated there was insufficient evidence that any of the 18 drugs evaluated provided significant benefits in fatigue (14).

Drug repurposing, in which existing compounds with known pharmacokinetic, pharmacodynamic, and safety profiles are identified with previously unsuspected novel activities, may provide an accelerated route to clinical translation. Given the high attrition rates in clinical trials for novel therapeutics identified and developed via traditional discovery pipelines, drug repurposing can provide significant advantages in terms of cost savings through the foreshortening of drug evaluation processes (15). Importantly, for pathologies such as ME/CFS, where the molecular basis has not been clearly established, repurposed drugs may provide insights into the underlying pathways and targets responsible.

While most repurposed drugs to date are the result of serendipitous discoveries, recent developments in computational methods allow disease transcriptomic signatures to be screened against extensive

databases of drug transcriptomic signatures for the identification of either positively or negatively correlated profiles (16). For example, positively correlated disease-drug transcriptomic profiles would suggest that the drug may induce a gene expression profile similar to that observed in the disease and so might enable insights into the mechanistic basis for pathogenesis. In contrast, negatively correlated disease-drug transcriptomic profiles would suggest that the drug may reverse the disease expression profile and so might have therapeutic potential in reversing the pathological phenotype. The large-scale transcriptomic profile database, Library of Integrated Cellular Signatures (LINCS) Connectivity Map (CMap L1000), contains over one million gene expression profiles regulated by over 40,000 genetic and small-molecule perturbagens, allowing disease expression signatures to be matched with negatively correlated drug-induced expression profiles for the identification of compounds with the potential to revert the pathological phenotype (17). Thus, LINCS CMap L1000 allows rational data-driven studies on drug repositioning based on signature reversion, for which there is proof-of-concept in a range of pathologies, including cancer and kidney disease (18, 19).

Given that pathological fatigue is an obligate symptom of ME/CFS for which there is currently no approved drug treatment, we have examined the gene expression profiles of ME/CFS patients from two independently published studies (20, 21). Our analysis identified dysregulated gene networks involved in immunity and metabolism that were common to both studies. Using CMap L1000, we identified compounds with known immune and metabolic targets with the potential to induce reciprocal profiles to those identified in ME/CFS patients. Focussing on compounds having known metabolic targets, we show that a peroxisome proliferator-activated receptor (PPAR) agonist (pioglitazone), a glycogen synthase kinase 3 (GSK3) inhibitor (tideglusib), and an agonist of mechanistic target of rapamycin complex 1 (mTORC1) were able to increase mitochondrial membrane potential and growth in C2C12 mouse muscle cells. These findings may provide new avenues for the development of therapies targeting cellular metabolism in long-term fatiguing conditions such as ME/CFS.

Methods

ME/CFS gene expression analysis

ME/CFS gene expression profiles were obtained from the publicly available GEO datasets GSE14577 and GSE227375 (20, 21). The GSE14577 dataset was obtained from the PBMCs of eight post-viral male ME/CFS patients and seven healthy male controls and was generated using Affymetrix Human Genome U133A and U133B Arrays (20). The GSE227375 dataset was obtained from the PBMCs of eight post-viral male ME/CFS patients and nine healthy male controls and was generated using Illumina NextSeq 500 RNA-seq technology (21). The GSE14577 dataset was parsed in GEOquery and then analysed using RStudio (v4.3.2) using the limma package. The GSE227375 dataset was analysed in RStudio using the DESeq2 package. The Benjamini and Hochberg method was applied for controlling the false discovery rate (FDR). Differentially expressed genes (DEGs) were analysed using the EnhancedVolcano package (v1.24.0). DEGs with an $FDR \leq 0.05$ and fold change (FC) ≥ 1.25 were considered significant.

Connectivity map (CMap) L1000 analysis

At the time of publication, the LINCS CMap L1000 database (<https://portals.broadinstitute.org/cmap/>) contained over one million gene expression profiles regulated by over 40,000 genetic and small-molecule perturbagens (17). We performed CMap L1000 analysis to identify novel i) disease-pathway relationships and ii) disease-drug relationships. The L1000 database was queried using the ME/CFS disease signature identified from GSE14577 and GSE227375 to identify perturbagens that induce i) similar gene expression profiles to ME/CFS (i.e., those with positive CMap scores) and ii) inverse gene expression profiles to ME/CFS (i.e., those with negative CMap scores). The identification of inverse disease-drug relationships in LINCS CMap L1000 allows the discovery of drugs with the potential to reverse a disease expression signature, which has been used to identify new therapeutics where *a priori* knowledge of disease mechanism, drug targets, and deregulated pathways may not be available (17, 22). Compounds with connectivity scores ≤ -0.64 , and $-\log_{10}(\text{FDR}) \geq 15$ were selected for further analysis.

Pathway and network analysis

The pathway enrichment analysis using KEGG and Reactome databases was performed in the ShinyGO (v0.81) tool, whereas for network analysis, we utilized STRING (v12.0). Gene networks were developed using STRING MCL clustering analysis.

Drug treatment of mouse C2C12 myoblasts and myotubes

The mouse C2C12 myoblast cell line (kind gift of Prof. Aaron Russell, Institute for Physical Activity and Nutrition, Deakin University) were maintained in medium (Dulbecco's Modified Eagle Medium [DMEM], 2mM L-glutamine, 100 units/ml penicillin and 0.1 mg/ml streptomycin) containing 10% heat-inactivated foetal bovine serum (HI-FBS)(all supplied by ThermoFisher Scientific) at 37°C in the presence of 5% CO₂ and passaged when < 80% confluent. For differentiation into myotubes, cells at > 80% confluency were washed in PBS and placed in medium (DMEM, 2mM L-glutamine, 100 units/ml penicillin and 0.1 mg/ml streptomycin) containing 1% horse serum (Sigma-Aldrich). The differentiation of C2C12 myoblasts was determined to be > 90% at 48 hours (h) by Hoechst 33342 (ThermoFisher Scientific) staining of multinucleated myotubes and confirmation of growth arrest using the MTT assay. Myotube cultures were replenished with fresh medium and 1% horse serum every 48h. Drugs utilized included pioglitazone (Jomar Life Research), rosiglitazone (Jomar Life Research), tideglusib (Jomar Life Research), and MHY1485 (Sigma-Aldrich). All drug compounds were dissolved in DMSO (Sigma–Aldrich) and subsequently serially diluted using sterile PBS (ThermoFisher Scientific) to the desired concentrations for treatments. For drug treatments, C2C12 myoblasts or myotubes were washed with PBS and then placed in either growth medium or differentiation medium, respectively, with drug titrations and time courses being performed.

Mitochondrial analysis

The mitochondrial membrane potential and mitochondrial mass were quantified using Tetramethylrhodamine Ethyl Ester (TMRE) (Jomar Life Research) and MitoTracker Green (MTG) (Sigma-Aldrich), respectively, according to the manufacturer's instructions. Triplicate wells containing either C2C12 myoblasts or myotubes were treated with either vehicle or the indicated concentrations of drugs (pioglitazone, rosiglitazone, tideglusib, and MHY1485) for up to 48h, after which 100mM TMRE or MTG was added for 30 minutes (min) at 37°C. TMRE fluorescence was analysed at 544nm (excitation) and 590nm (emission), and MTG fluorescence was analysed at 485nm (excitation) and 520nm (emission) using a microplate reader (BMG Labtech POLARstar Omega). For the analysis of intracellular ATP production, triplicate wells containing either C2C12 myoblasts or myotubes were treated with drugs for up to 48h, after which they were washed twice with PBS, and ATP was quantified by measuring bioluminescence using an ATP determination kit (ThermoFisher Scientific) following the manufacturer's instructions.

Reactive oxygen species (ROS) analysis

The intracellular ROS production in C2C12 myoblasts and myotubes was assessed using the MitoSOX Red (MSR) (ThermoFisher Scientific). Triplicate wells containing either C2C12 myoblasts or myotubes were treated with either vehicle or the indicated concentrations of drugs (pioglitazone, rosiglitazone, tideglusib, and MHY1485) for up to 60 minutes. Hydrogen peroxide (1mM) was included as a control for inducing mitochondrial ROS production. Cells were then incubated in 1µM MSR for up to 30 min before fluorescence was examined using a microplate reader (BMG Labtech POLARstar Omega) at 355nm (excitation) and 620nm (emission).

Cell growth analysis

The MTT (3-[4,5-dimethylthiazol-2-yl]-2,5 diphenyl tetrazolium bromide) assay (ThermoFisher Scientific) was carried out to measure C2C12 growth. C2C12 myoblasts or myotubes subjected to drug titrations were grown for up to 72h, following which MTT reagent was added for 3 h at 37°C. The medium was then removed, and solubilization buffer (1% w/v SDS, 0.01M HCL) was added. Absorbance signals at 550nm were analysed.

Results

Differential gene expression in ME/CFS

To identify dysregulated gene expression in ME/CFS, we analysed two independently published datasets in which gene expression in ME/CFS patients and healthy controls has been published (Fig. 1). In the first study, Gow *et. al.* (GEO Accession GSE14577) examined mRNA expression profiles using a microarray platform and reported DEGs involved in oxidative stress, apoptosis, and immune modulation

(20). In the second study, Gamer *et. al.* (GEO Accession GSE227375) examined gene expression using RNA-seq and reported DEGs involved in immune cell signalling, positive regulation of response to an external stimulus, and natural killer (NK) cell function (21).

We first examined both GSE14577 and GSE227375 to identify DEGs common to both datasets. The identified DEGs were then subjected to CMap L1000 analysis to identify i) functional gene networks and pathways involved in ME/CFS and ii) compounds with the potential to reverse ME/CFS disease signatures and so restore gene expression profiles to those observed in healthy control individuals. Using these CMap L1000 outputs, compounds were mapped to functional gene networks and prioritized for further examination for their *in vitro* activity in C2C12 muscle cells.

Volcano plots depicting DEGs are presented in Fig. 2A (GSE14577) and Fig. 2B (GSE227375). As expected, RNA-seq analysis resulted in larger fold changes (FC) in gene expression (Fig. 2B) compared to microarray analysis (Fig. 2A). A total of 312 significantly up-regulated and 286 significantly down-regulated DEGs were identified in the GSE14577 dataset, while 1243 significantly up-regulated and 1625 down-regulated DEGs were identified in the GSE227375 dataset (Fig. 2C, 2D; $FDR \leq 0.05$, $-0.3 < \text{Log}_2\text{FC} > 0.3$).

Table 1
Common differentially up-regulated genes between GSE14577 and GSE227375.

Gene Symbol	NCBI gene ID	Gene name	GSE14577		GSE227375	
			log2FC	Adjusted p-value	log2FC	Adjusted p-value
CALM3	808	Calmodulin 3	0.60	0.05	0.34	0.05
PGD	5226	Phosphogluconate dehydrogenase	0.55	0.05	0.55	0.001
RRAGD	58528	Ras related GTP binding D	0.55	0.04	0.67	0.01
NAAA	27163	N-acylethanolamine acid amidase	0.53	0.04	0.55	0.02
NAGA	4668	Alpha-N-acetylgalactosaminidase	0.52	0.04	0.32	0.02
PANX1	24145	Pannexin 1	0.51	0.03	0.57	0.05
XPNPEP1	7511	X-prolyl aminopeptidase 1	0.47	0.04	0.41	0.01
PTGS1	5742	Prostaglandin-endoperoxide synthase 1	0.43	0.04	0.93	0.005
SEN3-EIF4A1	100533955	SEN3-EIF4A1 readthrough	0.43	0.04	0.67	0.003
CANX	821	Calnexin	0.42	0.04	0.68	0.02
CDKN2D	1032	Cyclin dependent kinase inhibitor 2D	0.42	0.04	0.70	0.003
ZFAS1	441951	ZNF1 antisense RNA 1	0.40	0.05	0.86	0.04
FKBP15	23307	FKBP prolyl isomerase family member 15	0.38	0.03	0.42	0.01
ARPC4-TTLL3	100526693	ARPC4-TTLL3 readthrough	0.38	0.04	0.48	0.003
SUMO1	7341	Small ubiquitin like modifier 1	0.37	0.03	0.81	0.02
MIR1304	100302240	MicroRNA 1304	0.36	0.04	0.61	0.01
TGFBI	7045	Transforming growth factor beta induced	0.36	0.03	0.66	0.0003
BECN1	8678	Beclin 1	0.35	0.04	0.34	0.03
GNB4	59345	G protein subunit beta 4	0.35	0.04	0.89	0.002

Gene Symbol	NCBI gene ID	Gene name	GSE14577		GSE227375	
			log2FC	Adjusted p-value	log2FC	Adjusted p-value
MKRN1	23608	Makorin ring finger protein 1	0.34	0.04	0.57	0.01
RAP2C	57826	RAP2C, member of RAS oncogene family	0.33	0.05	0.82	0.04

Table 2
Common differentially downregulated genes between GSE14577 and GSE227375.

Gene Symbol	NCBI gene ID	Gene name	GSE14577		GSE227375	
			log2FC	Adjusted p-value	log2FC	Adjusted p-value
CBX5	23468	Chromobox 5	-0.53	0.03	-0.58	0.0003
ODF3B	440836	Outer dense fiber of sperm tails 3B	-0.52	0.02	-1.17	0.005
LOC102724965	102724965	Uncharacterized LOC102724965	-0.51	0.03	-1.25	0.0001
UQCC3	790955	Ubiquinol-cytochrome c reductase complex assembly factor 3	-0.49	0.01	-0.57	0.01
REXO1	57455	RNA exonuclease 1 homolog	-0.43	0.02	-1.55	0.0002
NATD1	256302	N-acetyltransferase domain containing 1	-0.42	0.03	-0.55	0.04
TRAPPC5	126003	Trafficking protein particle complex subunit 5	-0.42	0.03	-0.84	0.04
PWWP2B	170394	PWWP domain containing 2B	-0.41	0.04	-0.91	0.04
DPP7	29952	Dipeptidyl peptidase 7	-0.39	0.04	-0.98	0.0003
CAPS	828	Calcyphosine	-0.38	0.02	-1.13	0.0001
CACTIN	58509	Cactin, spliceosome C complex subunit	-0.37	0.02	-1.00	0.0005
ARRDC1-AS1	85026	ARRDC1 antisense RNA 1	-0.36	0.03	-0.60	0.004
KIF5A	3798	Kinesin family member 5A	-0.35	0.04	-1.07	0.03
ISYNA1	51477	Inositol-3-phosphate synthase 1	-0.34	0.04	-0.83	0.0002
BAIAP2L2	80115	BAR/IMD domain containing adaptor protein 2 like 2	-0.34	0.02	-1.14	0.01

Gene Symbol	NCBI gene ID	Gene name	GSE14577		GSE227375	
			log2FC	Adjusted p-value	log2FC	Adjusted p-value
WDR90	197335	WD repeat domain 90	-0.34	0.04	-1.15	0.002
C21orf58	54058	Chromosome 21 open reading frame 58	-0.33	0.02	-0.66	0.002
DDX54	79039	DEAD-box helicase 54	-0.33	0.04	-0.57	0.001
KCNC3	3748	Potassium voltage-gated channel subfamily C member 3	-0.33	0.05	-1.28	0.003
FRMD8	83786	FERM domain containing 8	-0.33	0.03	-0.43	0.02
LZTS2	84445	Leucine zipper tumor suppressor 2	-0.33	0.02	-0.87	0.01
SLC7A8	23428	Solute carrier family 7 member 8	-0.33	0.05	-1.65	0.01
LRP10	26020	LDL receptor related protein 10	-0.32	0.05	-0.67	0.001
ZFPM1	161882	Zinc finger protein, FOG family member 1	-0.32	0.05	-1.33	0.002
SETD5	55209	SET domain containing 5	-0.32	0.05	-0.48	0.02

Despite the heterogeneity between the studies of Gow *et. al.* (20) and Gamer *et. al.* (21) in terms of their methodologies, gene expression platforms, and patient cohorts, our analysis identified 21 up-regulated (Fig. 2C) and 25 down-regulated (Fig. 2D) DEGs that were common to both studies (Fig. 2C and 2D, Tables 1 and 2). This ME/CFS 46-gene signature included up-regulated DEGs known to encode proteins serving a broad range of cellular functions including calcium signalling, metabolism, G-protein signalling, ubiquitination and cell cycle regulation (Table 1). Additionally, down-regulated DEGs encode proteins involved in inositol lipid signalling, cell cycle regulation, RNA processing, and epigenetic regulation (Table 2).

ME/CFS pathway and network analysis

While KEGG and Reactome analyses of the 46 ME/CFS DEGs in Tables 1 and 2 identified a small number of generalized pathways that included homeostasis, dopaminergic neurons, and infection, insights into potential therapeutic targets and pathways from this analysis were limited (data not shown). The ability

of such tools to identify functional networks in disease is known to be limited when small panels of DEGs are analysed (e.g., our 46-gene signature), in part, because the results depend on the quality of the gene annotations and the completeness of the functional groups represented.

We therefore subjected the ME/CFS 46-gene signature to the CMap L1000 database that allows the identification of transcriptomic response similarities. The CMap L1000 database contains over one million mRNA signatures from genetic perturbagens (e.g., expression of shRNAs, cDNAs) in 244 cell lines (17). shRNA and cDNA perturbagens in the L1000 database that induce either similar or reciprocal gene expression profiles to those identified in a specific disease may provide insights into disease pathways and therapeutic targets. For example, the ability of specific shRNA and cDNA perturbagens in the L1000 platform to induce gene signatures similar (i.e., having positive Connectivity Scores) to those identified in Tables 1 and 2 would suggest that the targeted genes may, at least in part, control pathways that promote the ME/CFS clinical phenotype. Conversely, shRNA and cDNA perturbagens that induce reciprocal gene signatures (i.e., having negative Connectivity Scores) to those identified in Tables 1 and 2 would suggest that the targeted genes may have the potential to reverse the ME/CFS clinical phenotype (17).

Analysis of the ME/CFS 46-gene signature using CMap L1000 identified perturbagens targeting 83 genes (Supplementary Fig. 1, Supplementary Table 1). It is notable that, of the 83 genes having a $-\log_{10}\text{FDR} > 15$, all had negative connectivity scores less than -0.64 (Supplementary Fig. 1, inset). While it is not clear why the CMap L1000 analysis did not identify genes with $-\log_{10}\text{FDR} > 15$ with positive connectivity scores (Supplementary Fig. 1), it could be related to the proposal that ME/CFS patients demonstrate a hypometabolic state characterized by an overall decreased flux through multiple pathways (23).

We next performed STRING analysis on the 83 genes identified by CMap L1000 to gain insight into possible functional disease pathways. This analysis identified multiple networks involving i) cytokine/growth factor signalling, ii) infection and immunity, iii) metabolism and cell growth, iv) lipid metabolism, v) cytoskeletal regulation, vi) ubiquitination, vii) nucleotide metabolism, and viii) redox regulation (Fig. 3A). Pathway enrichment analysis using KEGG identified, i) metabolic pathways, ii) infection (Kaposi sarcoma associated herpes virus infection, Epstein-Barr virus infection, Yersinia infection), iii) cytokine signalling (TNF signalling), iv) growth factor signalling (ErbB, FoxO signalling), v) endocrine resistance, vi) reactive oxygen species (ROS) signalling and vii) autophagy pathways being represented within the gene network (Fig. 3B). Reactome analysis identified pathways involved in, i) metabolism, ii) immune response, iii) infection, iv) cytokine signalling, v) post-translational modifications and vi) cellular response to stress (Fig. 3C).

The identification of gene networks controlling cytokine signalling and infection in Fig. 3 is in line with reports that ME/CFS patients exhibit a range of immunological disruptions (24). In addition, multiple infectious pathogens identified in Fig. 3, including the SARS-CoV-2 virus responsible for COVID-19, have been linked with the ability to trigger ME/CFS (24, 25). Also prominent in our analysis was the identification of metabolic and cell growth pathways (Fig. 3).

CMap analysis of ME/CFS gene expression signatures for the identification of compounds.

The ME/CFS gene signature was then used to query the CMap L1000 database to identify compounds with the potential to induce reciprocal gene expression profiles (17). This approach allows the calculation of connectivity (similarity) scores between the ME/CFS DEGs and the gene signatures induced by 32,979 compounds across 244 cell lines. Compounds with negative connectivity scores are predicted to induce a gene expression profile that is negatively correlated with disease (17). Thus, compounds with negative connectivity scores may have the potential to reverse, at least in part, the gene expression profiles associated with ME/CFS and so may have therapeutic potential.

CMap L1000 analysis of the ME/CFS genes shown in Table 1 and Table 2 resulted in an output of compounds with connectivity scores ranging from - 0.73 to 0.72 (Fig. 4). From this analysis, a total of 14 compounds (0.01% of the total CMap L1000 database) with negative Connectivity Scores of ≤ -0.65 and $-\log_{10}(\text{FDR}) > 15.0$ were identified (Fig. 4B, inset). Consistent with our pathway analysis identifying gene networks controlling cytokine signalling, infection, and immunity in ME/CFS (Fig. 3), a total of 7 of the 14 (50%) compounds (7/14, 50%) were identified with known anti-inflammatory or analgesic activity (SB-705498, Mefenamic acid, Skimmianine, Ginkgolide-a, PF-750, PRE-084, and VX-702)(Supplementary Table 2). Also consistent with our pathway analysis (Fig. 3) identifying metabolic deregulation, 2/14 (14%) compounds (Supplementary Table 2) were identified with the ability to regulate mitochondrial metabolism or nutrient sensing (Timofibrate and Indoximod). In addition, 2/14 (14%) neurotransmitter regulators (Supplementary Table 2) used in the treatment of psychiatric illness were identified (Idalopirdine and Buspirone). Other compounds included a second mitochondrial-derived activator of caspases (SMAC) mimetic (GDC-0152), a histone deacetylase inhibitor (Pracinostat) used in the treatment of cancer, and a multi-targeted kinase inhibitor (Arcyriaflavin-a).

Although our analysis identified multiple compounds with anti-inflammatory and analgesic activity (Supplementary Table 2), clinical trials to date examining drugs targeting immune pathways in ME/CFS patients have not reported significant and sustained improvements for the majority of patients (8, 26–28). On the other hand, the identification of compounds using CMap L1000 with the ability to target energy metabolism was intriguing. ME/CFS patients have been reported to have significant reductions in aerobic metabolic capacity as measured in cardio-pulmonary exercise tests (CPET) (29, 30). Such impaired aerobic metabolism has been associated with altered blood metabolite profiles in patients that suggest impaired mitochondrial metabolism (23, 31). However, there are currently no effective pharmacological treatments targeting such defects associated with pathological fatigue in ME/CFS. We therefore focused our investigations on compound classes with the potential to regulate mitochondrial metabolism and nutrient sensing.

Indoximod decreases the conversion of tryptophan to N-formyl-kynurenine by the indoleamine 2,3-dioxygenase 1 (IDO1) enzyme. While the precise mechanism of action remains unclear, indoximod appears to i) reduce tryptophan levels, resulting in the induction of nutrient-sensing signals and the activation of mechanistic target of rapamycin complex 1 (mTORC1), and ii) induce the ligand-activated

transcription factor, aryl hydrocarbon receptor (AhR) involved in the expression of genes mediating immune responses (32). Thus, in addition to its ability to regulate mTORC1 metabolic pathways, indoximod can also regulate immune responses. In order to investigate the metabolic consequences of mTOR activation in the absence of immune transcription, we employed the mTORC1 agonist, MHY1485. In addition, because one of the consequences of GSK3 inhibition is the activation of mTORC1, we also included a GSK3 inhibitor, tideglusib, to further investigate perturbation of the mTOR pathway (33).

While timofibrate remains uncharacterized, the chemically-related fibrates, clofibrate and fenofibrate, are prodrugs that are converted to clofibric acid and fenofibric acid, respectively, in the liver and function as activators of PPAR α and PPAR γ (34). While fibrates such as fenofibrate are used clinically in the treatment of dyslipidemia associated with type 2 diabetes mellitus (T2DM) and metabolic syndrome, interpreting results from *in vitro* investigations, including C2C12 cells, can be difficult owing to cell-specific differences in the expression of the esterases necessary for prodrug conversion to active fenofibric acid (35). We therefore examined the more bioavailable PPAR agonists, pioglitazone and rosiglitazone (36).

Analysis of mitochondrial membrane potential, ATP production, and mitochondrial mass

We firstly investigated the impact of pioglitazone, rosiglitazone, tideglusib, and MHY1485 on mitochondrial metabolism in undifferentiated C2C12 myoblasts. While pioglitazone, rosiglitazone, and MHY1485 had no significant impact on mitochondrial membrane potential, ATP production, and mitochondrial mass, there was a modest but significant increase in both mitochondrial membrane potential (Fig. 5C) and ATP production (Fig. 5G) following treatment with tideglusib.

We further examined the effects of each compound on C2C12 differentiated myotubes. While neither pioglitazone nor rosiglitazone affected mitochondrial membrane potential or ATP production at 24h, 1nM pioglitazone significantly increased both mitochondrial membrane potential and ATP production at 48h (Fig. 6). Furthermore, 1nM of both Tideglusib and MHY1485 significantly increased mitochondrial membrane potential or ATP production at 24h, however, this increase was lost at 48h (Fig. 6). The increases in mitochondrial membrane potential (TMRE) or ATP production (luciferase) observed for pioglitazone, tideglusib, and MHY1485 were independent of mitochondrial mass suggesting that the response did not involve significant changes in mitochondrial biogenesis.

Analysis of ROS

In some cases, increased mitochondrial membrane potential and ATP production (as observed in Fig. 5, 6) can be associated with increased ROS production (37). We therefore examined ROS production in response to each compound. None of the four compounds significantly increased ROS production in C2C12 myoblasts (Supplementary Fig. 2A). However, there was a significant, albeit modest, transient increase in ROS production following 30 min of rosiglitazone and tideglusib in C2C12 myotubes (Supplementary Fig. 2B).

Analysis of C2C12 myoblast growth

We further examined the effects of the selected compounds on cell growth. While both 10nM pioglitazone and 10nM rosiglitazone increased cell growth of C2C12 myoblasts at 48h and 72h (Fig. 7A, 7B), no significant increases were observed for tideglusib or MHY1485 (Fig. 7C, 7D). Furthermore, none of the selected compounds had a significant effect on the growth of post-mitotic C2C12 myotubes (Supplementary Fig. 3).

Discussion

Using two independent transcriptomic studies reported by Gow *et. al.* and Gamer *et. al.*, we have identified a set of overlapping DEGs in ME/CFS patients from which we have investigated potential disease pathways and therapeutic avenues to treatment (20, 21). The overlapping gene expression signature identified genes involved multiple pathways broadly related to immune dysfunction, oxidative stress, apoptosis, and oxidative phosphorylation. Consistent with the post-infectious aetiology observed in some patients, immune dysfunction has been widely reported in ME/CFS (24, 25). However, despite such immune disruptions, the identification of reproducible cytokine or immune cell biomarkers in ME/CFS patients has so far remained elusive (38).

Interestingly, our analysis identified a number of common overlapping DEGs with proposed roles in metabolism including amino acid oxidation via the TCA cycle (SLC7a8, CAPS), oxidative stress (ARRDC1-AS1), mitochondrial ATP production (Panx1), mitochondrial biogenesis (RAP2C), metabolic reprogramming involving oxidative phosphorylation (UQCC3, CBX5), autophagy (BECN1) and the pentose phosphate pathway (PGD) suggesting disruptions in cellular metabolism (Tables 1 and 2). Further examination of the DEGs common to both datasets using the CMap L1000 database for gene-enrichment signatures (i.e., genetic perturbagens) allowed the identification of an 83-gene ME/CFS network (Supplementary Table 1). Given the post-infectious aetiology described for ME/CFS, it is perhaps not surprising that this network included multiple modules related to infection, cytokine signalling and immunity (Fig. 3). However, the identification of modules involving cellular metabolism and redox regulation in the 83-gene network suggests that such pathways may also be disrupted in ME/CFS (Fig. 3).

The molecular analysis of disease pathways has been widely employed to identify potential therapeutic targets from which lead compounds can be identified and optimized using preclinical models. However, for a range of chronic conditions such as ME/CFS, the lack of validated therapeutic targets, the unclear disease aetiology, together with the lack of proven preclinical models pose profound obstacles to drug discovery and validation. Within such a knowledge gap, therapeutic development of drugs for ME/CFS has been largely based on empirical clinical trials in which existing drugs have been repurposed based on clinical rationales rather than preclinical mechanistic evidence. We therefore examined an alternative approach that does not require *a priori* knowledge of preclinical mechanism or disease aetiology in which compounds are identified with the potential to induce gene signatures that are reciprocal (or

inverse) to those observed in ME/CFS patients using the CMap L1000 platform (17). Such an approach has been widely employed for the identification of compounds that have the potential to reverse disease gene expression profiles and so could provide a novel therapeutic route for drug development (17–19, 22).

Consistent with our pathway mapping (Fig. 3), the majority of compounds identified using the CMap L1000 database have known indications related to immunity, inflammation, and pain (Supplementary Table 2, SB-705498, Mefenamic acid, Skimmianine, Indoximod, Ginkgolide-a, PRE-084, and VX-702). However, despite the consistent association between immune dysfunction and ME/CFS, clinical trials investigating immunotherapies such as intravenous immunoglobulin G, anti-histamines, hydrocortisone and fludrocortisone, IL-1 antagonism, and B-cell ablation using rituximab have not, so far, demonstrated sustained responses in the majority of patients (8, 26–28). One possible exception may be cyclophosphamide, an alkylating agent widely used in cancer treatment, which was found to result in significant symptom improvement and remissions in some patients (39). In addition, the significant association of viral infections preceding the onset of ME/CFS has led to a number of clinical trials investigating antiviral therapies, including rintatolimod and valganciclovir, with some significant improvements (40, 41). However, the safety profiles and variable therapeutic windows of such drugs remain an issue in terms of clinical translation.

While ME/CFS is associated with a range of autonomic impairments, our pathway analysis did not identify neurological networks (Fig. 3). However, our CMap L1000 analysis identified drugs known to modify neurotransmission in psychiatric conditions. For example, buspirone (Supplementary Table 2) modifies serotonin and dopamine neurotransmission and is used clinically in the treatment of anxiety and depression. Idalopirdine (Supplementary Table 2) modifies serotonin, glutamate, and cholinergic neurotransmission and is used in the treatment of dementia and schizophrenia. While antidepressants, including buspirone, have been used in the treatment of ME/CFS, their efficacy remains unclear (42). For example, despite the reported involvement of the dopamine and serotonin pathways in ME/CFS, trials examining ondansetron, duloxetine, phenelzine, and fluoxetine have not identified significant improvements in patients (8, 28).

Our analysis identifying disruption of metabolic pathways in ME/CFS together the CMap L1000 identification of compounds known to be metabolic regulators, suggested that the targeting of cellular metabolism may provide one therapeutic avenue for the treatment of ME/CFS. In examining the metabolic PPAR agonists, pioglitazone and rosiglitazone, only pioglitazone significantly increased mitochondrial membrane potential and ATP production in C2C12 myotubes (Fig. 6). This increase did not appear to be associated with the induction of mitochondrial biogenesis as we did not observe any change in mitochondrial mass (Fig. 6). Furthermore, despite the increased mitochondrial membrane potential and ATP production in response to pioglitazone, we did not observe increased ROS production (Supplementary Fig. 2). While pioglitazone and rosiglitazone are PPAR γ agonists, they are also able to activate PPAR α at higher concentrations (36). It is noteworthy that pioglitazone is nearly 10-times less potent as an activator for PPAR γ than rosiglitazone (36). Thus, it is possible that the ability of

pioglitazone but not rosiglitazone to increase mitochondrial membrane potential in C2C12 myotubes may be due to the activation of PPAR α rather than PPAR γ . While such a proposal would be consistent with the reported increase in PPAR α mRNA expression following C2C12 myotube differentiation, further work will be required to determine the mechanistic basis for the difference between pioglitazone and rosiglitazone observed in our studies (43).

Pioglitazone and rosiglitazone are approved clinically in the treatment of T2DM. While their ability to activate PPAR-mediated transcriptional regulation of metabolic genes such as Glucose Transporter Type 4, adiponectin and Acyl-CoA Synthetase Long Chain Family Member 1 are thought to underpin improvements in insulin sensitivity, there is evidence that they can also improve mitochondrial metabolism (44). For example, pioglitazone has been shown to enhance oxygen consumption and ATP production in insulin-resistant C2C12 muscle cells (45). In part, the ability of PPAR agonists to increase mitochondrial metabolism may be responsible for the increased growth capacity of C2C12 myoblasts observed in Fig. 7. Others have shown the decrease in exercise capacity, ADP-dependent mitochondrial respiration, complex I and III activities, and citrate synthase activity observed in high-fat diet diabetic mice were all ameliorated following pioglitazone treatment (46). While it remains unclear whether pioglitazone and rosiglitazone improve mitochondrial metabolism solely via PPAR-dependent pathways, the discovery that pioglitazone can directly bind the MitoNEET protein to regulate oxidative capacity suggests that multiple pathways may be involved (47).

The underlying molecular basis for the pathological fatigue experienced by ME/CFS patients remains unclear. However, several lines of evidence suggest that impaired mitochondrial function may be a contributing factor. Primary peripheral blood monocyctic cells, primary skeletal muscle cells and immortalized lymphocytes from ME/CFS patients exhibit a reduced mitochondrial coupling efficiency, ATP synthesis and oxygen consumption rate compared to cells isolated from healthy controls (48, 49). Furthermore, Mandarano *et al.* found a significant reduction in mitochondrial membrane potential in primary CD8⁺ T cells isolated from ME/CFS patients (50). The reduced mitochondrial metabolism in ME/CFS is proposed to result in altered metabolic profiles whereby compensatory anaplerosis in which specific amino acids and metabolites are used to replenish TCA intermediates thereby bridging the aerobic deficit (31). Additionally, Naviaux *et al.* have shown that ME/CFS patients have abnormal metabolite profiles in their peripheral blood that are consistent with an overall reduction in metabolic capacity (23). Thus, given our findings that pioglitazone is able to increase mitochondrial membrane potential in C2C12 myoblasts, future work will be important for determining whether it can restore mitochondrial metabolism, including oxygen consumption and ATP production, in primary cells isolated from ME/CFS patients.

To our knowledge, the ability of pioglitazone to improve pathological fatigue in patients has not been reported. However, there is evidence that pioglitazone can improve skeletal muscle metabolism *in vivo*. For example, pioglitazone significantly improved oxygen consumption, anaerobic threshold and skeletal muscle fatty acid oxidation in a small cohort of metabolic syndrome patients subjected to bicycle ergometer exercise testing (51). Furthermore, while oxygen consumption in skeletal muscle explants

taken from T2DM patients was reduced compared to age- and BMI-matched control subjects, explants taken from T2DM patients that had been treated with pioglitazone (but not rosiglitazone) demonstrated partial restoration (52). In line with the observed impairment in mitochondrial metabolism in cells isolated from ME/CFS patients, several studies have reported reduced oxygen consumption and anaerobic thresholds in patients when measured by cardio-pulmonary exercise tests (29, 30). Thus, future work examining the ability of pioglitazone to increase mitochondrial metabolism including oxidative phosphorylation in primary cells obtained from ME/CFS patients may pave the way for future clinical trials in which the re-purposing of pioglitazone for the treatment of ME/CFS could be considered.

Our results further showed that tideglusib induced a transient increase of mitochondrial membrane potential and ATP production in both C2C12 myoblasts and myotubes (Fig. 5, 6). The increased mitochondrial membrane potential in response to tideglusib did not appear to be due to mitochondrial biogenesis, as mitochondrial mass remained unchanged (Fig. 5, 6). Tideglusib is a GSK-3 β inhibitor that is currently in clinical development for degenerative pathologies, including Alzheimer's disease and myotonic dystrophy (53, 54). While such conditions are known to be associated with protein aggregation and proteasomal dysfunction, they are also characterized by impaired mitochondrial metabolism. For example, in a study by Martin *et. al.*, GSK3 β inhibition both in cell lines and in mice increased mitochondrial membrane potential and respiration (55). Importantly, these studies found that GSK3 β inhibition increased the stability of PPAR coactivator 1-alpha (PGC-1 α) that is a key regulator of the PPARs (55). In part, such a pathway may facilitate the ability of PGC1 α to increase aspartate and glutamate levels, leading to improved energy utilization and mitochondrial respiration (56). Also consistent with our findings, Theeuwes *et. al.* found that siRNA inhibition of GSK3 β in C2C12 cells increased mitochondrial respiration (57). Tideglusib has been shown to improve oxidative metabolism and reduce muscle fatigue in a C57BL/10-mdx mouse model of muscular dystrophy (58). Similarly, tideglusib demonstrated improvements in neuromuscular symptoms, including fatigue, in a Phase II study of myotonic dystrophy type 1, although whether the improvement was mediated via mitochondrial metabolism was not assessed (54). Given its favourable pharmacokinetic and toxicity profiles, further investigation into the potential repurposing of tideglusib for the treatment of ME/CFS is warranted (54).

Crosstalk has been reported for GSK3 and mTORC1 in a number of cell types whereby GSK3 inhibition can lead to mTORC1 activation and, reciprocally, mTORC1 activation can lead to GSK3 inactivation (59). Given our findings with the GSK3 inhibitor, tideglusib, we also analysed the mTORC1 activator, MHY1485. Similar to tideglusib, MHY1485 was also shown to transiently increase mitochondrial membrane potential and ATP production in C2C12 myotubes in a manner that was independent of both mitochondrial mass and ROS production (Fig. 6, Supplementary Fig. 2). Although both mTORC1 and GSK3 are known to control cell growth and metabolism, neither MHY1485 nor tideglusib significantly affected the growth of C2C12 myoblasts (Fig. 7). While our studies did not investigate the mechanisms by which MHY1485 may influence mitochondrial membrane potential, Bantug *et. al.* reported that mTORC signalling via GSK3 at mitochondria-ER contact sites promotes the TCA cycle flux that is important for the effector functions of memory CD8⁺ T cells suggesting functional roles in mitochondrial respiration (60). In addition, while the over-expression of the IRX5 homeobox transcription factor

disrupted mitochondrial structure and impeded mitochondrial oxygen consumption in human bone marrow-derived mesenchymal stem cells, these effects could be partly overcome by MHY1485, implicating a role for mTORC1 activation in mitochondrial function (61). It could be that the impaired mitochondrial metabolism in ME/CFS is due to the accumulation of dysfunctional mitochondria as a consequence of aberrant autophagy. BECN1 (Beclin-1), a key regulator of autophagy that integrates signals from mTORC1, was identified as part of our ME/CFS disease signature (Table 1). Others have reported that the autophagy-related protein, ATG13, is upregulated in the serum of ME/CFS patients, suggesting impaired or abortive autophagy (62). While approaches that promote mTORC1 activity could conceivably restore defective autophagic pathways, it is noteworthy that a single-arm clinical trial in which ME/CFS patients were treated with a low dose of the mTORC1 inhibitor, rapamycin, reported improvements in multiple symptoms, including fatigue and PEM, although the improvements did not always correlate with autophagic markers such as BECN1 (63). Further work will be important in determining whether the targeting of mTORC1 may be therapeutically beneficial in ME/CFS.

Our study has several limitations. Most notably, our study was performed in a single C2C12 muscle cell line, and so there will be important differences in terms of the *in vivo* physiology in ME/CFS patients. In addition, while the application of CMap has had some success in identifying candidate drugs for repurposing in a number of pathologies, the drug-induced gene expression profiles within CMap were not generated from cells derived from ME/CFS patients. Thus, the ability of a drug to revert an ME/CFS disease signature in CMap cell lines may not translate to reversion in ME/CFS patients. Furthermore, while we have examined short-term drug effects on mitochondrial metabolism and growth, we have not examined the effects of each compound on the expression of ME/CFS disease signature genes. Future work will be important in examining primary cells derived from ME/CFS patients and the ability of each compound to revert the disease signature. Lastly, determinants of therapeutic efficacy are complex and involve a range of factors beyond matching inverse compound-disease expression profiles. For example, therapeutic responses can be determined by the bioavailability of the drug to target tissues and its therapeutic window. Thus, further preclinical investigations will be important before randomized prospective ME/CFS clinical trials could be considered.

Despite over four decades of research, ME/CFS remains a disease for which there is no clearly understood mechanistic basis, no reproducible biomarkers, no clear therapeutic targets and no validated pre-clinical models. Developing new drug discovery pipelines within such a 'knowledge vacuum' presents substantial barriers to clinical translation. These issues are compounded by the escalating costs and length of time required to deliver new therapies following conventional drug development pipelines that involve target identification, lead development, *in vitro* evaluation, pre-clinical testing and clinical trials. Even if such conventional pipelines identify a new drug with therapeutic potential, the current risk-adverse economic climate and high failure rates can hinder development. For ME/CFS, an off-the-shelf approach has predominated in which the majority of controlled drug trials have examined existing immunotherapies that were selected on the basis that ME/CFS is an immune disease (64). Although such *quasi ad hoc* approaches are important for orphan diseases such as ME/CFS, alternative approaches in which drug repurposing based on the unbiased matching of reciprocal disease:drug gene

expression signatures may offer important alternative routes to clinical trials. To date, reciprocal matching of disease:drug expression signatures has been applied to a number of diseases with some success (16, 18, 19). Thus, with a detailed understanding of ME/CFS aetiology, biomarkers, therapeutic targets, and pre-clinical models still lacking, rational approaches to drug repurposing, such as described herein, may be an attractive alternative to conventional drug development pathways.

To our knowledge, there are no approved drug therapies targeting pathological fatigue associated with mitochondrial dysfunction. In the case of ME/CFS, while some dietary and nutritional supplements aimed at improving mitochondrial function have been investigated, objective improvements have not been observed (65). With pathological fatigue being a prevalent symptom associated with diverse diseases, 'metabolic' drugs may have wide application provided that both efficacy and safety can be demonstrated. In particular, with the finding that long-COVID demonstrates many of the clinical hallmarks of ME/CFS including pathological fatigue, the approaches outlined herein may have wider application, particularly for diseases for which traditional drug development pathways may not be available and for which there is urgent unmet need (25).

Abbreviations

ME/CFS

Myalgic Encephalomyelitis/Chronic Fatigue Syndrome

DEG

Differentially expressed gene

MS

Multiple sclerosis

CRF

Cancer-related fatigue

PSF

Post-stroke fatigue

LINCS

Library of Integrated Cellular Signatures

CMap

Connectivity Map

PPAR

Peroxisome proliferator-activated receptor

GSK3

Glycogen synthase kinase 3

mTORC1

Mechanistic target of rapamycin complex 1

GEO

Gene expression omnibus

FDR

False discovery rate
FC
Fold change
DMEM
Dulbecco's Modified Eagle Medium
HI-FBS
Heat-inactivated foetal bovine serum
TMRE
Tetramethylrhodamine Ethyl Ester
MTG
MitoTracker Green
MSR
MitoSOX Red
ROS
Reactive oxygen species
SMAC
Second mitochondrial-derived activator of caspases
CPET
Cardio-pulmonary exercise tests
T2DM
Type 2 diabetes mellitus
PGC-1 α
PPAR coactivator 1-alpha

Declarations

Ethical approval and consent to participate

Not applicable

Consent for Publication

Not applicable

Availability of data and materials

All data generated or analysed during this study are included in this published article and its supplementary information files.

Competing interests

The authors declare no conflict of interest.

Funding

The authors received no financial support for the research.

Author's contributions

AD: Conceptualization, Data curation, formal analysis, investigation, methodology, project administration, software, validation, visualization, Writing – Original Draft Preparation, Writing – Review & Editing

MAG: Conceptualization, Data curation, funding acquisition, investigation, methodology, project administration, resources, supervision, validation, Writing – Original Draft Preparation, Writing – Review & Editing

ZA: Data curation, formal analysis, investigation, methodology, validation, visualization, Writing – Review & Editing

Acknowledgements

We thank all researchers who have publicly shared gene expression data, enabling open, reproducible, and impactful scientific research.

References

1. Nijrolder I, van der Windt DA, van der Horst HE. Prognosis of fatigue and functioning in primary care: a 1-year follow-up study. *Ann Fam Med*. 2008;6(6):519–27.
2. Cullen W, Kearney Y, Bury G. Prevalence of fatigue in general practice. *Ir J Med Sci*. 2002;171(1):10–2.
3. Louati K, Berenbaum F. Fatigue in chronic inflammation - a link to pain pathways. *Arthritis Res Ther*. 2015;17:254.
4. Falk Hvidberg M, Brinth LS, Olesen AV, Petersen KD, Ehlers L. The Health-Related Quality of Life for Patients with Myalgic Encephalomyelitis / Chronic Fatigue Syndrome (ME/CFS). *PLoS ONE*. 2015;10(7):e0132421.
5. Sprangers MA, de Regt EB, Andries F, van Agt HM, Bijl RV, de Boer JB, et al. Which chronic conditions are associated with better or poorer quality of life? *J Clin Epidemiol*. 2000;53(9):895–907.
6. Stadjé R, Dornieden K, Baum E, Becker A, Biroga T, Bosner S, et al. The differential diagnosis of tiredness: a systematic review. *BMC Fam Pract*. 2016;17(1):147.
7. Matura LA, Malone S, Jaime-Lara R, Riegel B. A Systematic Review of Biological Mechanisms of Fatigue in Chronic Illness. *Biol Res Nurs*. 2018;20(4):410–21.
8. Kim DY, Lee JS, Park SY, Kim SJ, Son CG. Systematic review of randomized controlled trials for chronic fatigue syndrome/myalgic encephalomyelitis (CFS/ME). *J translational Med*. 2020;18(1):7.

9. Smith ME, Haney E, McDonagh M, Pappas M, Daeges M, Wasson N, et al. Treatment of Myalgic Encephalomyelitis/Chronic Fatigue Syndrome: A Systematic Review for a National Institutes of Health Pathways to Prevention Workshop. *Ann Intern Med.* 2015;162(12):841–50.
10. Perez DQ, Espiritu AI, Jamora RDG. Efficacy and safety of amantadine for the treatment of fatigue in multiple sclerosis: a systematic review and meta-analysis. *Neurodegener Dis Manag.* 2020;10(6):383–95.
11. Shangyan H, Kuiqing L, Yumin X, Jie C, Weixiong L. Meta-analysis of the efficacy of modafinil versus placebo in the treatment of multiple sclerosis fatigue. *Mult Scler Relat Disord.* 2018;19:85–9.
12. Yennurajalingam S, Lu Z, Rozman De Moraes A, Tull NN, Kubiak MJ, Geng Y et al. Meta-Analysis of Pharmacological, Nutraceutical and Phytopharmaceutical Interventions for the Treatment of Cancer Related Fatigue. *Cancers (Basel).* 2022;15(1).
13. Chu SH, Zhao X, Komber A, Cheyne J, Wu S, Cowey E, et al. Systematic review: Pharmacological interventions for the treatment of post-stroke fatigue. *Int J Stroke.* 2023;18(9):1071–83.
14. Mucke M, Mochamat, Cuhls H, Peuckmann-Post V, Minton O, Stone P et al. Pharmacological treatments for fatigue associated with palliative care. *Cochrane Database Syst Rev.* 2015(5):CD006788.
15. Sun D, Gao W, Hu H, Zhou S. Why 90% of clinical drug development fails and how to improve it? *Acta Pharm Sinica B.* 2022.
16. Pushpakom S, Iorio F, Eyers PA, Escott KJ, Hopper S, Wells A, et al. Drug repurposing: progress, challenges and recommendations. *Nat Rev Drug Discov.* 2019;18(1):41–58.
17. Subramanian A, Narayan R, Corsello SM, Peck DD, Natoli TE, Lu X, et al. A Next Generation Connectivity Map: L1000 Platform and the First 1,000,000 Profiles. *Cell.* 2017;171(6):1437–e5217.
18. Ramsey JM, Kettyle LM, Sharpe DJ, Mulgrew NM, Dickson GJ, Bijl JJ, et al. Entinostat prevents leukemia maintenance in a collaborating oncogene-dependent model of cytogenetically normal acute myeloid leukemia. *Stem Cells.* 2013;31(7):1434–45.
19. Klein J, Caubet C, Camus M, Makridakis M, Denis C, Gilet M, et al. Connectivity mapping of glomerular proteins identifies dimethylaminoparthenolide as a new inhibitor of diabetic kidney disease. *Sci Rep.* 2020;10(1):14898.
20. Gow JW, Hagan S, Herzyk P, Cannon C, Behan PO, Chaudhuri A. A gene signature for post-infectious chronic fatigue syndrome. *BMC Med Genomics.* 2009;2:38.
21. Gamer J, Van Booven DJ, Zarnowski O, Arango S, Elias M, Kurian A et al. Sex-Dependent Transcriptional Changes in Response to Stress in Patients with Myalgic Encephalomyelitis/Chronic Fatigue Syndrome: A Pilot Project. *Int J Mol Sci.* 2023;24(12).
22. Lamb J, Crawford ED, Peck D, Modell JW, Blat IC, Wrobel MJ, et al. The Connectivity Map: using gene-expression signatures to connect small molecules, genes, and disease. *Science.* 2006;313(5795):1929–35.
23. Naviaux RK, Naviaux JC, Li K, Bright AT, Alaynick WA, Wang L, et al. Metabolic features of chronic fatigue syndrome. *Proc Natl Acad Sci USA.* 2016;113(37):E5472–80.

24. Bansal AS, Bradley AS, Bishop KN, Kiani-Alikhan S, Ford B. Chronic fatigue syndrome, the immune system and viral infection. *Brain Behav Immun*. 2012;26(1):24–31.
25. Dehlia A, Guthridge MA. The persistence of myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) after SARS-CoV-2 infection: A systematic review and meta-analysis. *J Infect*. 2024;89(6):106297.
26. Fluge O, Rekeland IG, Lien K, Thurmer H, Borchgrevink PC, Schafer C, et al. B-Lymphocyte Depletion in Patients With Myalgic Encephalomyelitis/Chronic Fatigue Syndrome: A Randomized, Double-Blind, Placebo-Controlled Trial. *Ann Intern Med*. 2019;170(9):585–93.
27. Roerink ME, Bredie SJH, Heijnen M, Dinarello CA, Knoop H, Van der Meer JWM. Cytokine Inhibition in Patients With Chronic Fatigue Syndrome: A Randomized Trial. *Ann Intern Med*. 2017;166(8):557–64.
28. Seton KA, Espejo-Oltra JA, Gimenez-Orenga K, Haagmans R, Ramadan DJ, Mehlsen J et al. Advancing Research and Treatment: An Overview of Clinical Trials in Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS) and Future Perspectives. *J Clin Med*. 2024;13(2).
29. Stevens S, Snell C, Stevens J, Keller B, VanNess JM. Cardiopulmonary Exercise Test Methodology for Assessing Exertion Intolerance in Myalgic Encephalomyelitis/Chronic Fatigue Syndrome. *Front Pead*. 2018;6:242.
30. Keller BA, Pryor JL, Giloteaux L. Inability of myalgic encephalomyelitis/chronic fatigue syndrome patients to reproduce VO₂ peak indicates functional impairment. *J translational Med*. 2014;12:104.
31. Fluge O, Mella O, Bruland O, Risa K, Dyrstad SE, Alme K, et al. Metabolic profiling indicates impaired pyruvate dehydrogenase function in myalgic encephalopathy/chronic fatigue syndrome. *JCI insight*. 2016;1(21):e89376.
32. Brincks EL, Adams J, Wang L, Turner B, Marcinowicz A, Ke J, et al. Indoximod opposes the immunosuppressive effects mediated by IDO and TDO via modulation of AhR function and activation of mTORC1. *Oncotarget*. 2020;11(25):2438–61.
33. Souder DC, Anderson RM. An expanding GSK3 network: implications for aging research. *Geroscience*. 2019;41(4):369–82.
34. Li HX, Sun MR, Zhang Y, Song LL, Zhang F, Song YQ, et al. Human Carboxylesterase 1A Plays a Predominant Role in Hydrolysis of the Anti-Dyslipidemia Agent Fenofibrate in Humans. *Drug Metab Dispos*. 2023;51(11):1490–8.
35. Murakami R, Murakami H, Kataoka H, Cheng XW, Takahashi R, Numaguchi Y, et al. Unmetabolized fenofibrate, but not fenofibric acid, activates AMPK and inhibits the expression of phosphoenolpyruvate carboxykinase in hepatocytes. *Life Sci*. 2010;87(15–16):495–500.
36. Sakamoto J, Kimura H, Moriyama S, Odaka H, Momose Y, Sugiyama Y, et al. Activation of human peroxisome proliferator-activated receptor (PPAR) subtypes by pioglitazone. *Biochem Biophys Res Commun*. 2000;278(3):704–11.
37. Lebieczinska-Arciszewska M, Suski J, Bonora M, Pakula B, Pinton P, Duszynski J, et al. The Relation Between Mitochondrial Membrane Potential and Reactive Oxygen Species Formation. *Methods Mol*

38. Corbitt M, Eaton-Fitch N, Staines D, Cabanas H, Marshall-Gradisnik S. A systematic review of cytokines in chronic fatigue syndrome/myalgic encephalomyelitis/systemic exertion intolerance disease (CFS/ME/SEID). *BMC Neurol.* 2019;19(1):207.
39. Rekeland IG, Fosså A, Lande A, Ktoridou-Valen I, Sørland K, Holsen M et al. Intravenous Cyclophosphamide in Myalgic Encephalomyelitis/Chronic Fatigue Syndrome. An Open-Label Phase II Study. *Front Med.* 2020;7.
40. Watt T, Oberfoell S, Balise R, Lunn MR, Kar AK, Merrihew L, et al. Response to valganciclovir in chronic fatigue syndrome patients with human herpesvirus 6 and Epstein-Barr virus IgG antibody titers. *J Med Virol.* 2012;84(12):1967–74.
41. Strayer DR, Young D, Mitchell WM. Effect of disease duration in a randomized Phase III trial of rintatolimod, an immune modulator for Myalgic Encephalomyelitis/Chronic Fatigue Syndrome. *PLoS ONE.* 2020;15(10):e0240403.
42. Leong KH, Yip HT, Kuo CF, Tsai SY. Treatments of chronic fatigue syndrome and its debilitating comorbidities: a 12-year population-based study. *J translational Med.* 2022;20(1):268.
43. Shimizu Y, Hamada K, Guo T, Hasegawa C, Kuga Y, Takeda K, et al. Role of PPARalpha in inflammatory response of C2C12 myotubes. *Biochem Biophys Res Commun.* 2024;694:149413.
44. Fang L, Zhang M, Li Y, Liu Y, Cui Q, Wang N. PPARgene: A Database of Experimentally Verified and Computationally Predicted PPAR Target Genes. *PPAR Res.* 2016;2016:6042162.
45. Pagel-Langenickel I, Bao J, Joseph JJ, Schwartz DR, Mantell BS, Xu X, et al. PGC-1alpha integrates insulin signaling, mitochondrial regulation, and bioenergetic function in skeletal muscle. *J Biol Chem.* 2008;283(33):22464–72.
46. Takada S, Hirabayashi K, Kinugawa S, Yokota T, Matsushima S, Suga T, et al. Pioglitazone ameliorates the lowered exercise capacity and impaired mitochondrial function of the skeletal muscle in type 2 diabetic mice. *Eur J Pharmacol.* 2014;740:690–6.
47. Wiley SE, Murphy AN, Ross SA, van der Geer P, Dixon JE. MitoNEET is an iron-containing outer mitochondrial membrane protein that regulates oxidative capacity. *Proc Natl Acad Sci USA.* 2007;104(13):5318–23.
48. Tomas C, Brown AE, Newton JL, Elson JL. Mitochondrial complex activity in permeabilised cells of chronic fatigue syndrome patients using two cell types. *PeerJ.* 2019;7:e6500.
49. Missailidis D, Annesley SJ, Allan CY, Sanislav O, Lidbury BA, Lewis DP et al. An Isolated Complex V Inefficiency and Dysregulated Mitochondrial Function in Immortalized Lymphocytes from ME/CFS Patients. *Int J Mol Sci.* 2020;21(3).
50. Mandarano AH, Maya J, Giloteaux L, Peterson DL, Maynard M, Gottschalk CG et al. Myalgic encephalomyelitis/chronic fatigue syndrome patients exhibit altered T cell metabolism and cytokine associations. *J Clin Invest.* 2020.
51. Yokota T, Kinugawa S, Hirabayashi K, Suga T, Takada S, Omokawa M, et al. Pioglitazone improves whole-body aerobic capacity and skeletal muscle energy metabolism in patients with metabolic

- syndrome. *J Diabetes Investig.* 2017;8(4):535–41.
52. Rabol R, Boushel R, Almdal T, Hansen CN, Ploug T, Haugaard SB, et al. Opposite effects of pioglitazone and rosiglitazone on mitochondrial respiration in skeletal muscle of patients with type 2 diabetes. *Diabetes Obes Metab.* 2010;12(9):806–14.
53. Lovestone S, Boada M, Dubois B, Hull M, Rinne JO, Huppertz HJ, et al. A phase II trial of tideglusib in Alzheimer's disease. *J Alzheimers Dis.* 2015;45(1):75–88.
54. Horrigan J, Gomes TB, Snape M, Nikolenko N, McMorn A, Evans S, et al. A Phase 2 Study of AMO-02 (Tideglusib) in Congenital and Childhood-Onset Myotonic Dystrophy Type 1 (DM1). *Pediatr Neurol.* 2020;112:84–93.
55. Martin SA, Souder DC, Miller KN, Clark JP, Sagar AK, Eliceiri KW, et al. GSK3 β Regulates Brain Energy Metabolism. *Cell Rep.* 2018;23(7):1922–31. e4.
56. Agudelo LZ, Ferreira DMS, Dadvar S, Cervenka I, Ketscher L, Izadi M, et al. Skeletal muscle PGC-1 α reroutes kynurenine metabolism to increase energy efficiency and fatigue-resistance. *Nat Commun.* 2019;10(1):2767.
57. Theeuwes WF, Gosker HR, Langen RCJ, Pansters NAM, Schols A, Remels AHV. Inactivation of glycogen synthase kinase 3 β (GSK-3 β) enhances mitochondrial biogenesis during myogenesis. *Biochim Biophys Acta Mol Basis Dis.* 2018;1864(9 Pt B):2913–26.
58. Marcella BM, Hockey BL, Braun JL, Whitley KC, Geromella MS, Baranowski RW, et al. GSK3 inhibition improves skeletal muscle function and whole-body metabolism in male mouse models of Duchenne muscular dystrophy. *Nat Commun.* 2024;15(1):10210.
59. Evangelisti C, Chiarini F, Paganelli F, Marmioli S, Martelli AM. Crosstalks of GSK3 signaling with the mTOR network and effects on targeted therapy of cancer. *Biochim Biophys Acta Mol Cell Res.* 2020;1867(4):118635.
60. Bantug GR, Fischer M, Grahlert J, Balmer ML, Unterstab G, Develioglu L, et al. Mitochondria-Endoplasmic Reticulum Contact Sites Function as Immunometabolic Hubs that Orchestrate the Rapid Recall Response of Memory CD8(+) T Cells. *Immunity.* 2018;48(3):542–55. e6.
61. Jiang B, Zheng J, Yao H, Wang Y, Song F, Huang C. IRX5 suppresses osteogenic differentiation of hBMSCs by inhibiting protein synthesis. *J Cell Physiol.* 2024;239(6):e31286.
62. Gottschalk G, Peterson D, Knox K, Maynard M, Whelan RJ, Roy A. Elevated ATG13 in serum of patients with ME/CFS stimulates oxidative stress response in microglial cells via activation of receptor for advanced glycation end products (RAGE). *Mol Cell Neurosci.* 2022;120:103731.
63. Ruan BT, Bulbule S, Gile B, Reyes A, Chheda B, Bateman L, et al. Low-dose rapamycin alleviates clinical symptoms of fatigue and PEM in ME/CFS patients via improvement of autophagy: a pilot study. *J translational Med.* 2025;23(1):1148.
64. Whiting P, Bagnall AM, Sowden AJ, Cornell JE, Mulrow CD, Ramirez G. Interventions for the treatment and management of chronic fatigue syndrome: a systematic review. *JAMA.* 2001;286(11):1360–8.
65. Campagnolo N, Johnston S, Collatz A, Staines D, Marshall-Gradisnik S. Dietary and nutrition interventions for the therapeutic treatment of chronic fatigue syndrome/myalgic encephalomyelitis:

Figures

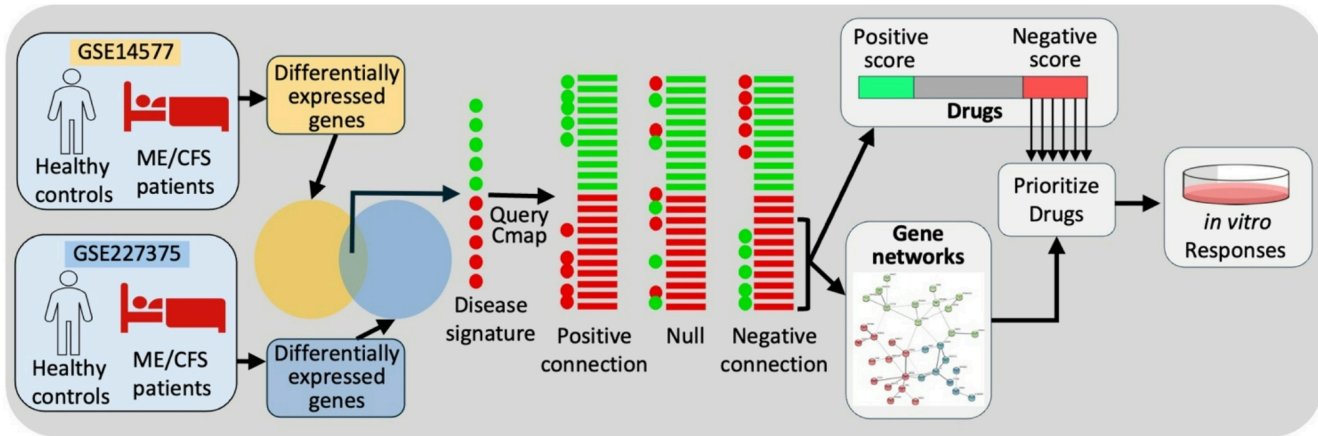


Figure 1

Workflow. Drug repurposing scheme for diseases with unclear aetiology and no approved therapies. The differentially expressed genes (DEGs) were identified using two independent datasets (GSE14577 and GSE227375). DEGs common to both datasets were subjected to Connectivity Map (CMAP) L1000 for the identification of functional gene networks and potential therapeutic compounds for ME/CFS. The compounds were further prioritized based on their targets and tested *in vitro* for their ability to improve mitochondrial bioenergetics.

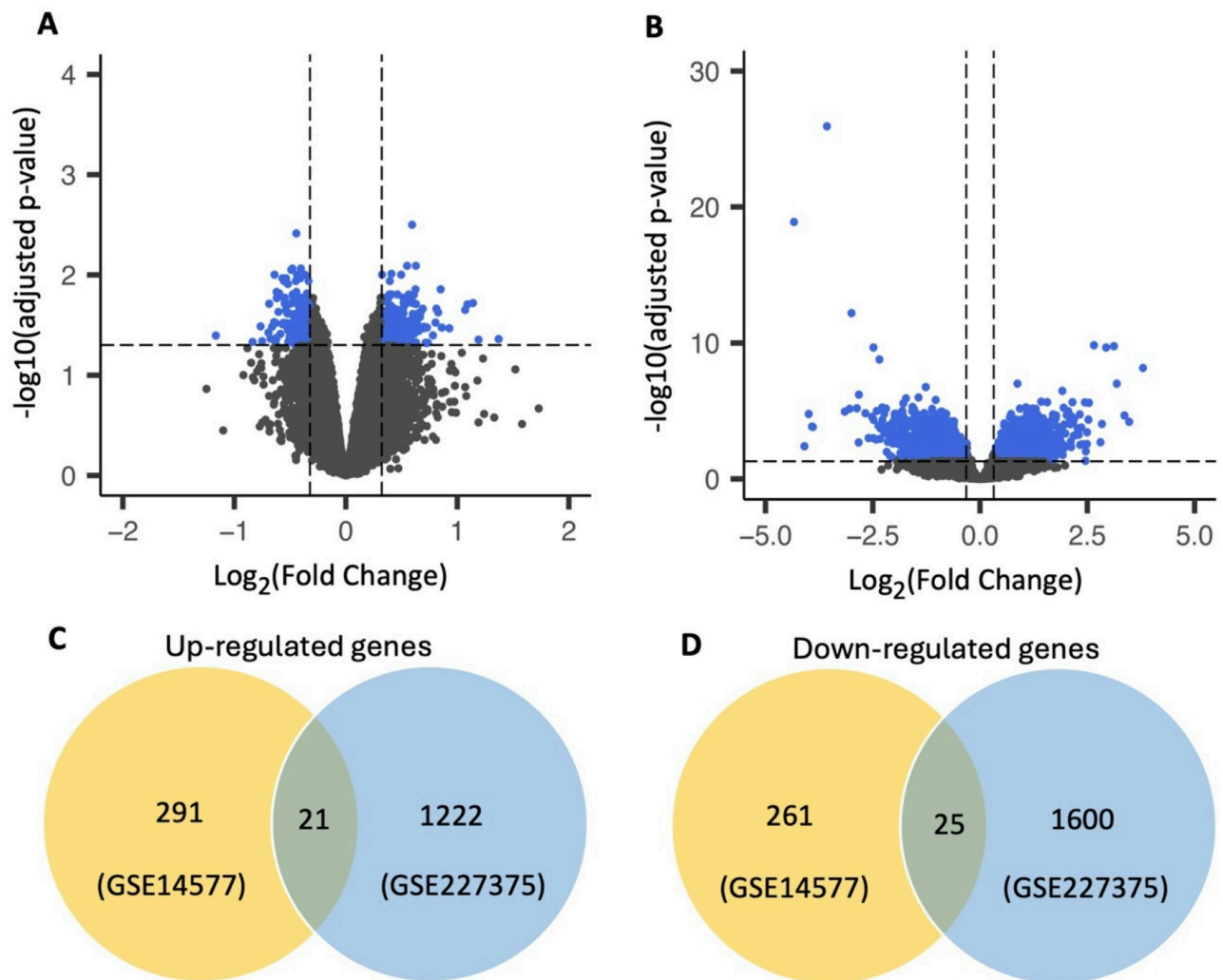


Figure 2

***in-silico* identification of DEGs in ME/CFS.** The microarray GSE14577 and RNAseq GSE227375 datasets were analysed to identify DEGs. **A)** Volcano plot of the microarray GSE14577 dataset, and **B)** Volcano plot of the RNAseq GSE227375 dataset. The data were log transformed, and significantly DEGs were determined using $FDR \leq 0.05$ and $|\log_2(FC)| \geq 0.32$. The blue dots represent significantly expressed genes, whereas the grey dots represent non-significant genes. **C)** Venn diagram showing common up-regulated genes between microarray and RNAseq datasets. **D)** Venn diagram showing common down-regulated genes between microarray and RNAseq datasets. The details of these common genes are provided in Table 1 and Table 2.

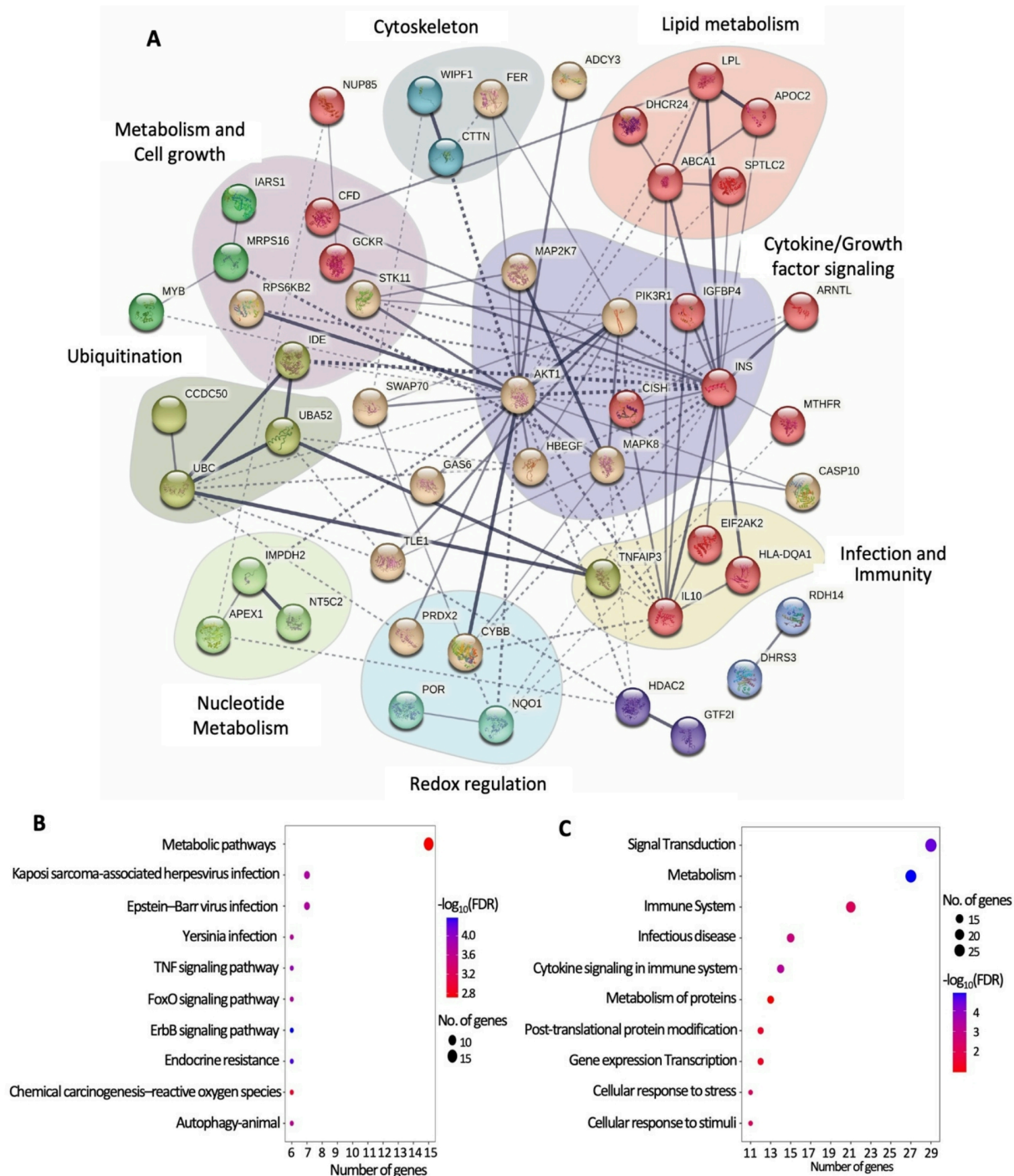


Figure 3

Gene networks and pathway enrichment in ME/CFS. A) The 83 genes with negative connectivity scores were subjected to identify gene networks using STRING (v12.0) with an interaction score of 0.4. The MCL clustering with inflation of 3 was applied to the full gene network. The major gene networks identified were cytokine/growth factor signalling, metabolism and cell growth, lipid metabolism, infection and immunity, redox regulation, cytoskeleton, ubiquitination, and nucleotide metabolism. **B)** The KEGG

pathways and **C**)Reactome pathways were analysed using 83 genes with negative connectivity score in ShinyGO (v0.82), and graphs were plotted using the *ggplot2* package in R.

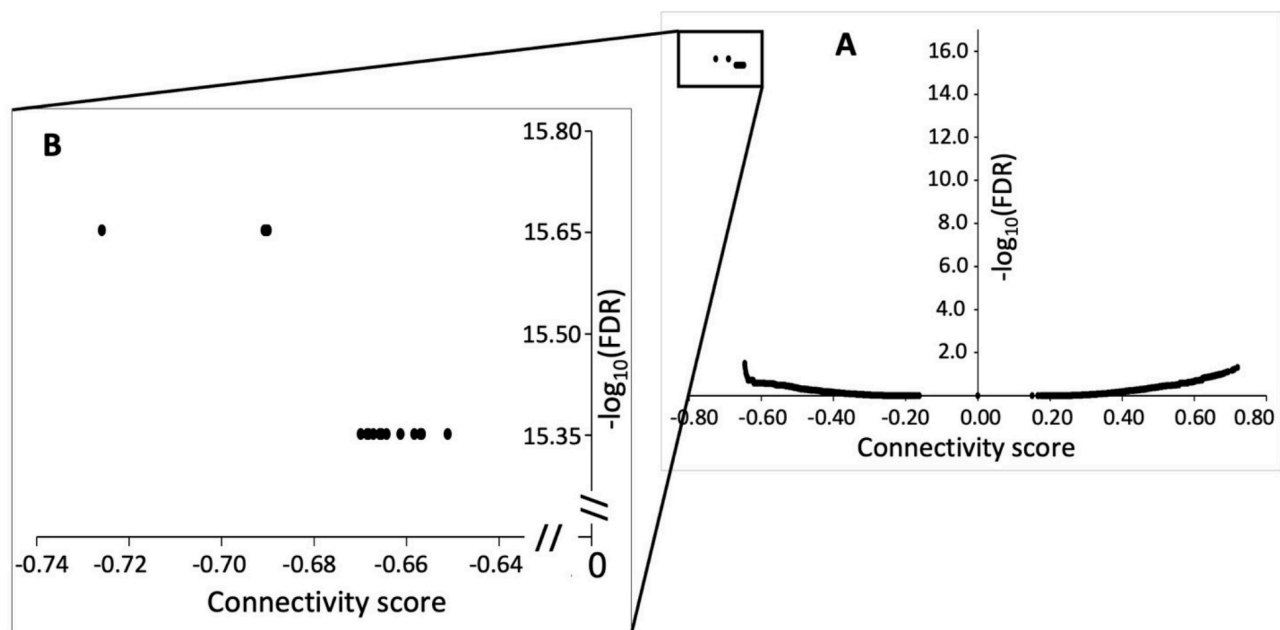


Figure 4

Drugs identified by CMap. The common DEGs were queried in CMap, and drugs were identified. Each point in the graph represents a drug plotted against the individual connectivity score and $-\log_{10}\text{FDR}$. The drugs with the highest negative connectivity score have the potential to be repurposed for ME/CFS. The boxed inset region of **(A)** is expanded to show top-ranked compounds with connectivity scores < -0.65 and $-\log_{10}(\text{FDR}) > 15.35$ **(B)**.

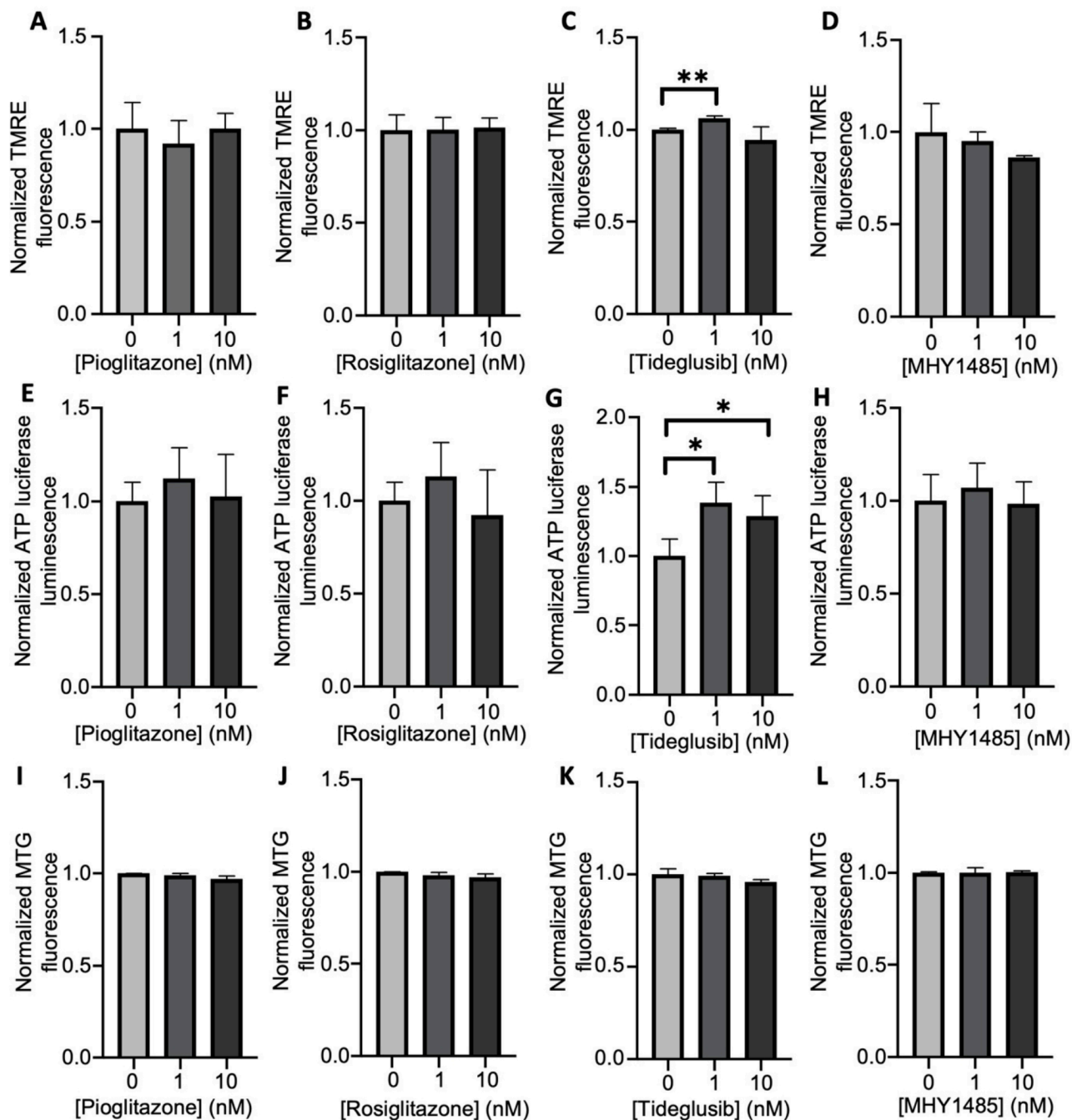


Figure 5

The effect of Pioglitazone, Rosiglitazone, Tideglusib, and MHY1485 on mitochondrial membrane potential, ATP production, and mitochondrial biogenesis in C2C12 myoblasts. The C2C12 myoblasts were treated with either vehicle (0 nM), 1 nM, or 10 nM of the indicated compounds. The mitochondrial membrane potential (**A-D**), ATP production (**E-H**) and mitochondrial biogenesis (**I-L**) were estimated using TMRE, ATP luciferase, and MTG assay, respectively, at 24 hours. Values shown as mean $\pm$ standard

deviation. *p*-values were calculated using an unpaired t-test. Significant differences are denoted with asterisks * *p*-value <0.05 and ** *p*-value <0.01.

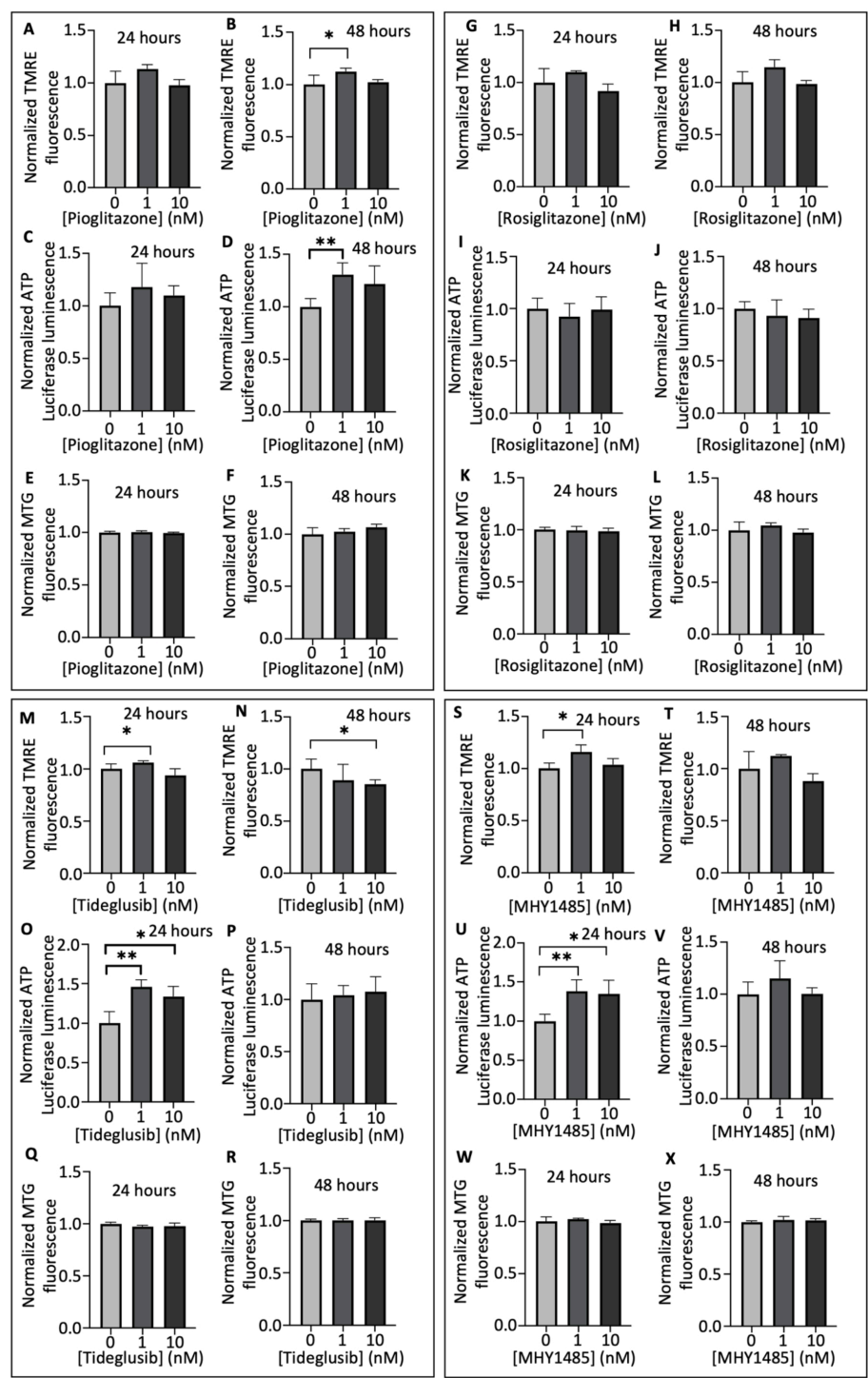


Figure 6

The effect of Pioglitazone, Rosiglitazone, Tideglusib, and MHY1485 on mitochondrial membrane potential, ATP production, and mitochondrial biogenesis in C2C12 myotubes. The C2C12 myotubes

were treated with either vehicle (0 nM), 1 nM, or 10 nM of the indicated compounds. The mitochondrial membrane potential, ATP production, and mitochondrial biogenesis were estimated using TMRE, ATP luciferase, and MTG assay, respectively, over 48 hours. Values shown as mean $\pm$ standard deviation. *P*-values were calculated using an unpaired t-test. Significant differences are denoted with asterisks * *p*-value <0.05 and ** *p*-value <0.01.

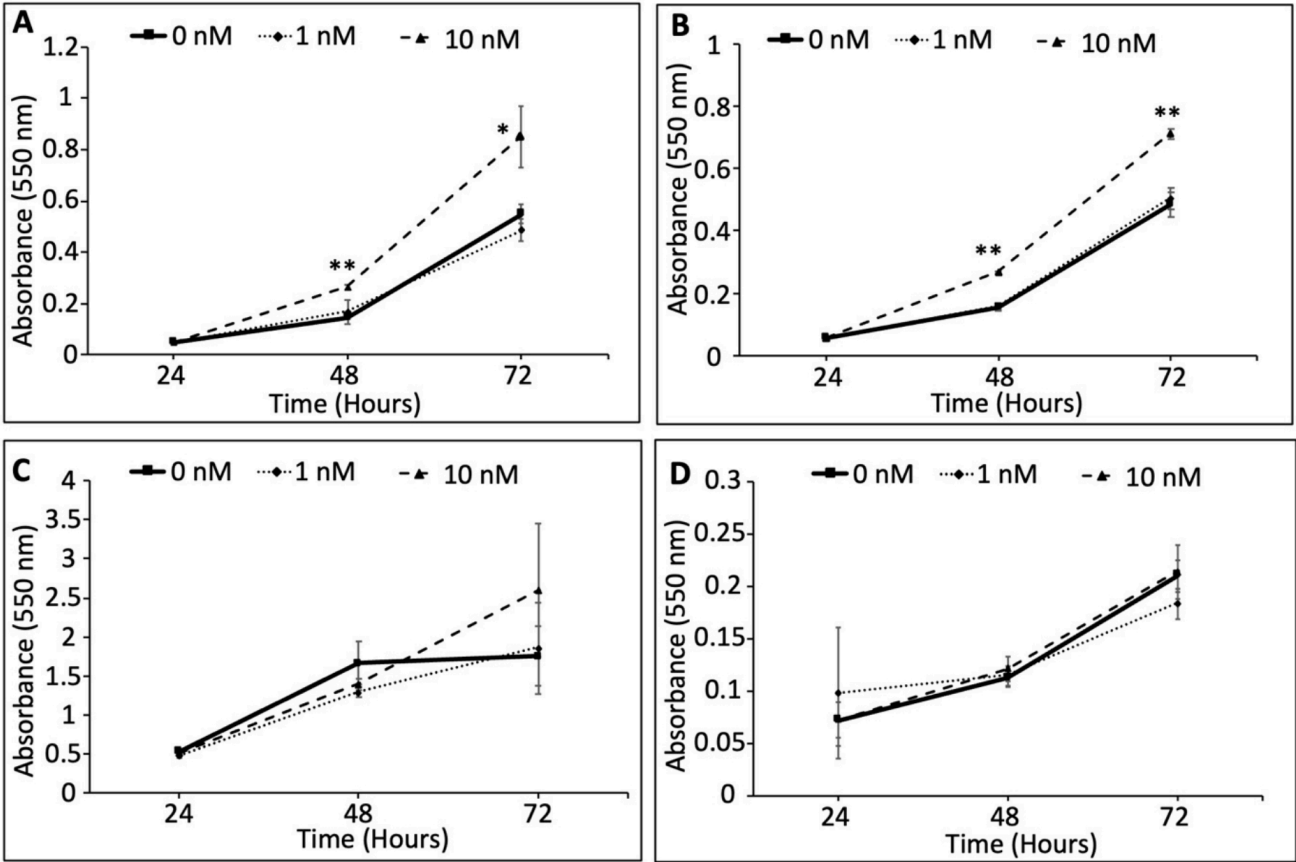


Figure 7

The effect of A) Pioglitazone, B) Rosiglitazone, C) Tideglusib, and D) MHY1485 on cell growth. The C2C12 myoblasts were treated with either vehicle (0 nM), 1 nM, or 10 nM of the indicated compounds, and cell growth was measured using the MTT assay over 72 hours. Values shown as mean $\pm$ standard deviation. *P*-values were calculated using an unpaired t-test. Significant differences are denoted with asterisks * *p*-value <0.05 and ** *p*-value <0.01.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [Supplementarydata.pdf](#)